Lecture for 10th week

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Statically Indeterminate Problems

1.General Considerations

- If the equations of statics suffice (足够) for determining the external reactions as well as the internal stress resultants, a structural system is statically determinate. Otherwise, it is statically indeterminate.

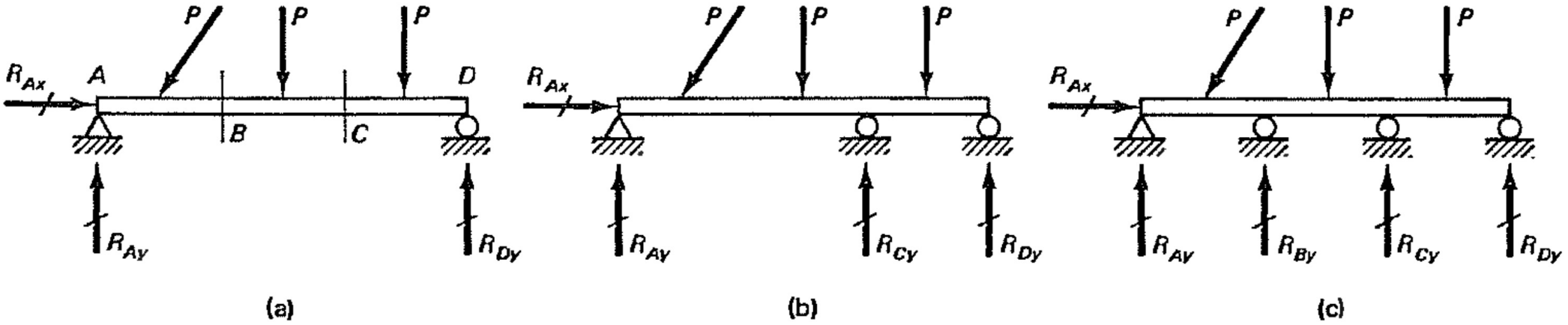


Fig. 1-16 Identical beam with identical loading having different support conditions

1.General Considerations

- The reactions that can be removed leaving a stable system statically determinate are redundant. Depending on the number (n) of the redundant internal forces or reactions, the system is said to be indeterminate to the n-th degree.

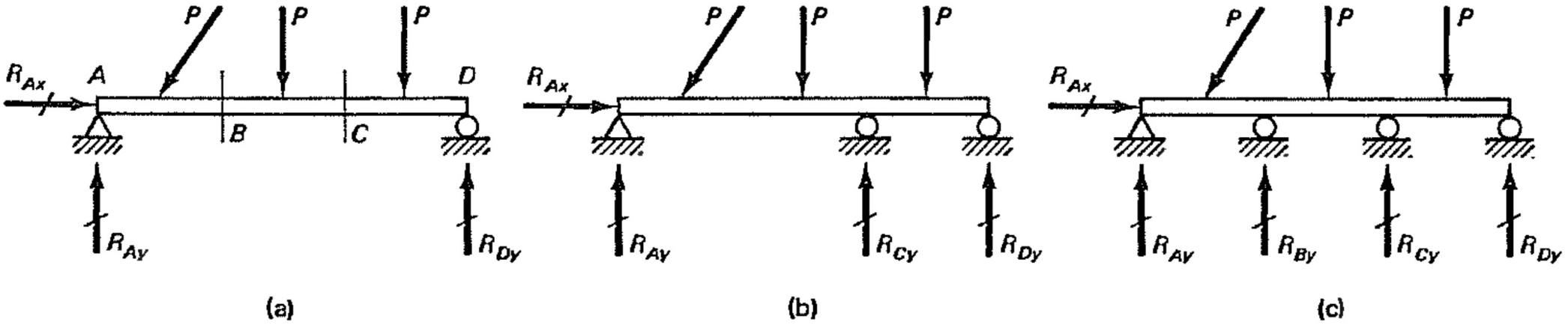
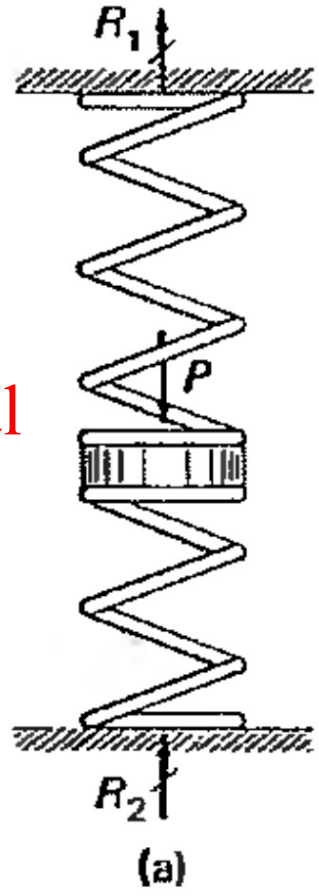


Fig. 1-16 Identical beam with identical loading having different support conditions

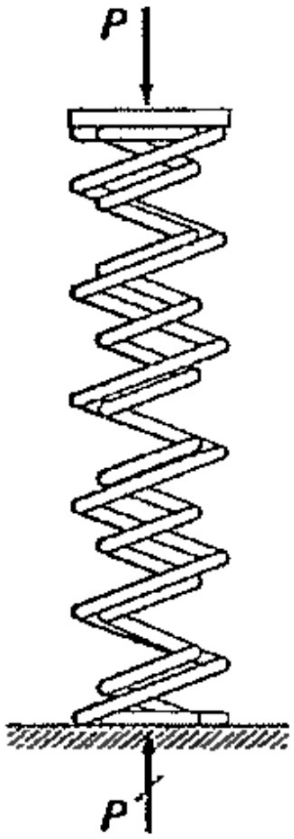
1.General Considerations

external statical indeterminacy



(a)

internal statical indeterminacy



(b)

For both instances, the deformation characteristics of the system components must be considered.

1.General Considerations

- Reduce a problem to statical determinacy such that the internal forces can readily be found. Common to all these procedures, the same three basic concepts encountered before are applied, and must be satisfied:

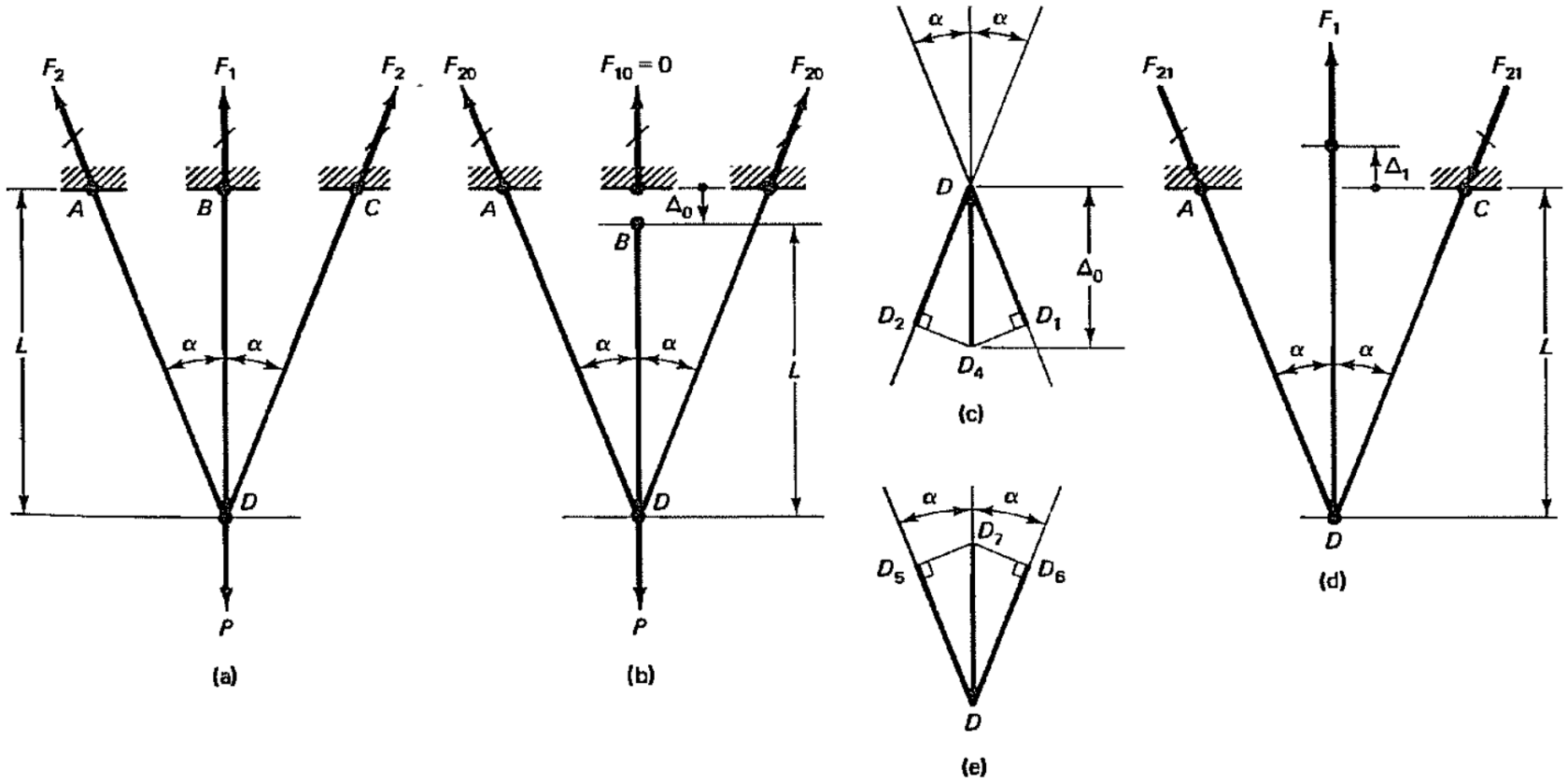
1. Equilibrium conditions for the system must be assured both in the local and global sense.
2. Geometric compatibility among the deformed parts of a body and at the boundaries must be satisfied.
3. Constitutive relations (stress-strain relations) for the materials of the system must be complied with.

1. General Considerations

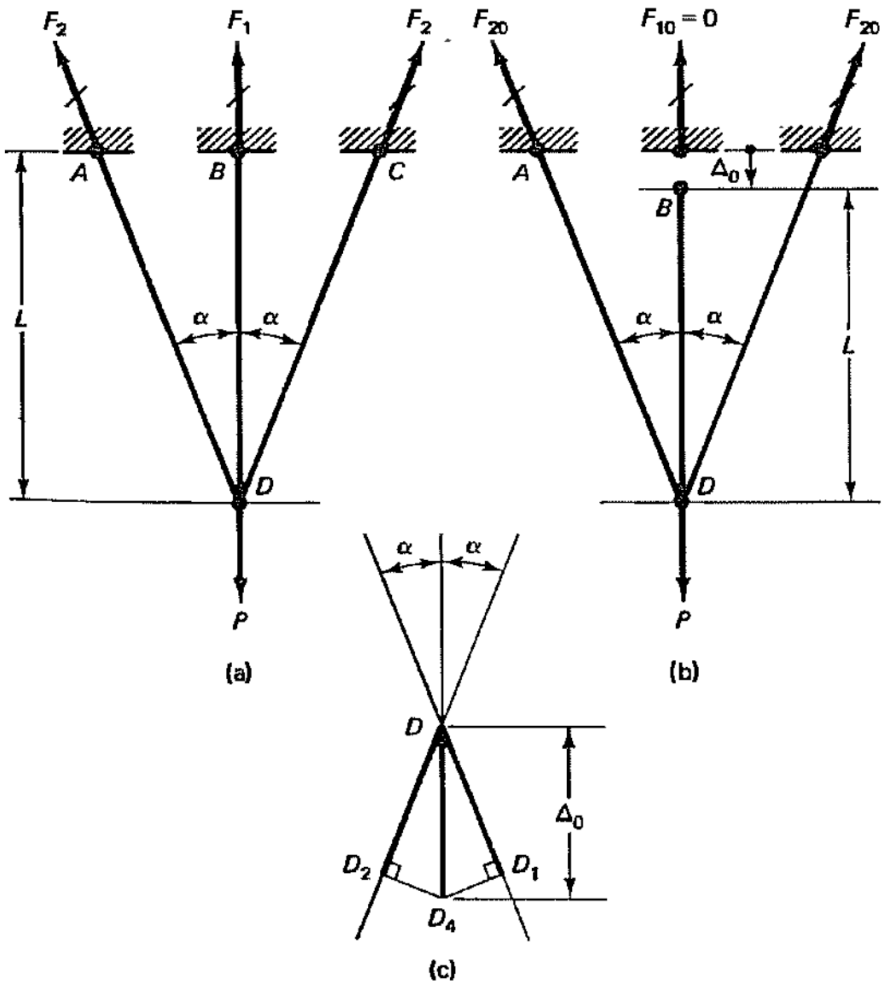
- **Force Method**: First removing and then restoring redundant reactions such that the compatibility condition at boundaries is satisfied. Solution is obtained directly for the unknown reaction forces.
- **Displacement Method**: The compatibility of displacements of adjoining members and at the boundaries is maintained throughout the loading process, and solution for displacements are obtained from equilibrium equations.

2. Force Method of Analysis - example 4

For the planar system of the three elastic bars shown in Fig. 2-47(a), determine the forces in the bars caused by applied force P. The cross-sectional area A of each bar is the same, and their elastic modulus is E.



2. Force Method of Analysis - example 4



$$F_{10} = 0 \qquad \text{and} \qquad 2F_{20}\cos\alpha = P$$

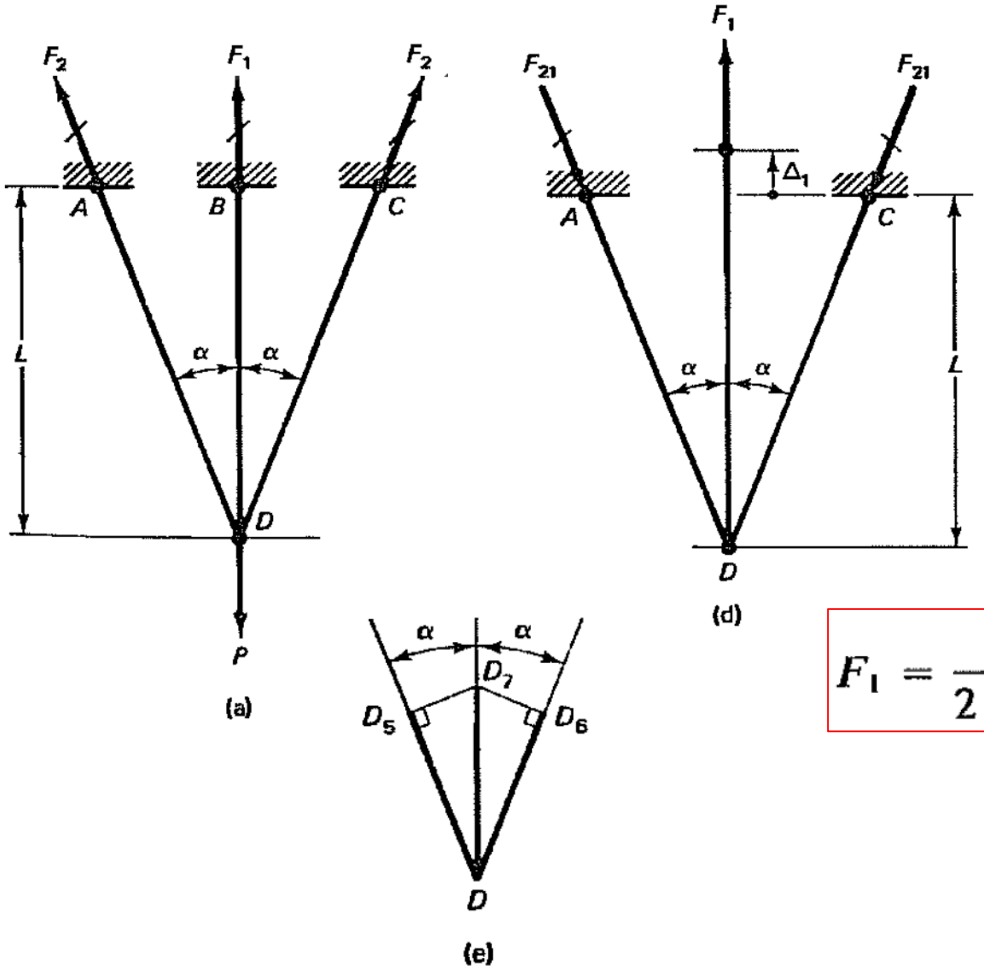
$$F_{20} = \frac{P}{2\cos\alpha}$$

$$L_{AD}\cos\alpha = L, \qquad L_{AD} = L/\cos\alpha$$

$$(\Delta_{AD})_0 = \frac{PL}{2AE\cos^2\alpha}$$

$$\Delta_0\cos\alpha = (\Delta_{AD})_0 \qquad \text{and} \qquad \Delta_0 = -\frac{PL}{2AE\cos^3\alpha}$$

2. Force Method of Analysis - example 4



$$\Delta_1 = \frac{F_1 L}{AE} + \frac{F_1 L}{2AE \cos^3 \alpha}$$

$$\Delta_0 + \Delta_1 = 0,$$

$$F_1 = \frac{P}{2 \cos^3 \alpha + 1} \quad \text{and} \quad F_2 = \frac{P}{2 \cos^3 \alpha + 1} \cos^2 \alpha$$

2. Force Method of Analysis - Comments

The algebraic sum of the two solutions is an application of the principle of superposition. This principle is based upon the premise (前提) that the resultant stress or strain in system due to several forces is the algebraic sum of their effects when separately applied. This assumption is true only if each effect is linearly related to the force causing it.

3. Displacement Method of Analysis

- **Displacement Method**: The compatibility of displacements of adjoining members and at the boundaries is maintained throughout the loading process, and solution for displacements are obtained from equilibrium equations.

3. Displacement Method of Analysis

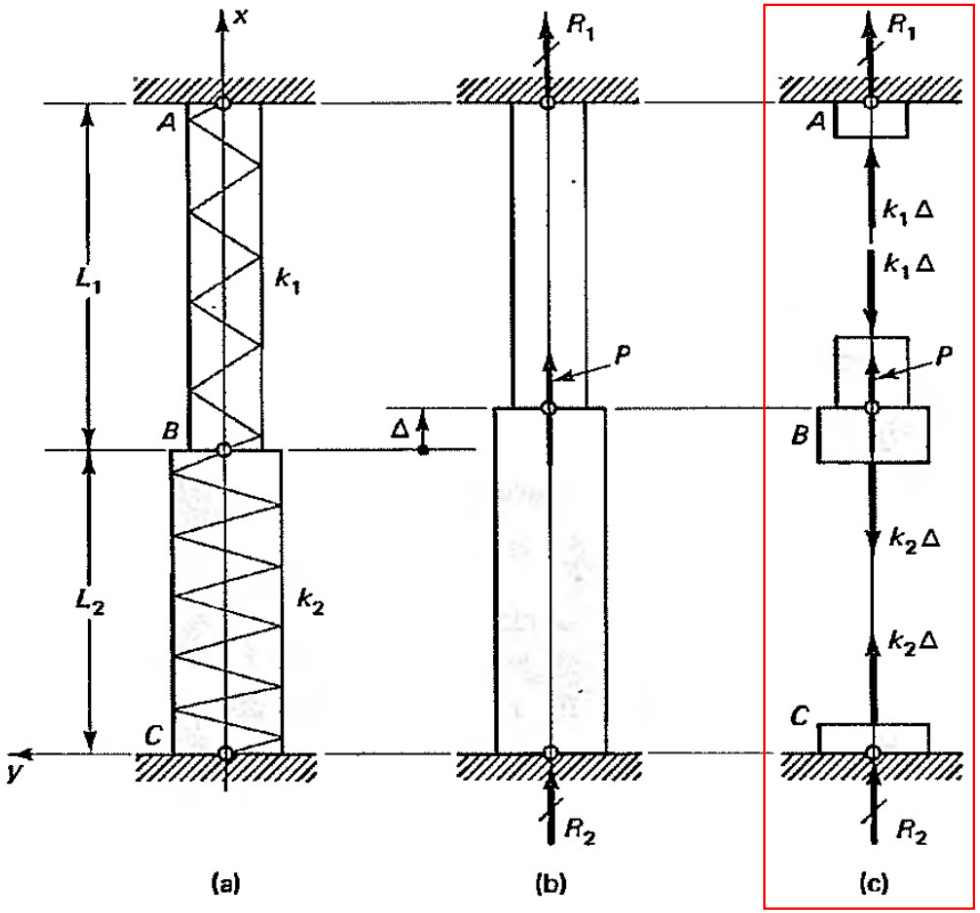


Fig. 2-48 Displacement (stiffness) method of analysis for a statically indeterminate axially loaded bar.

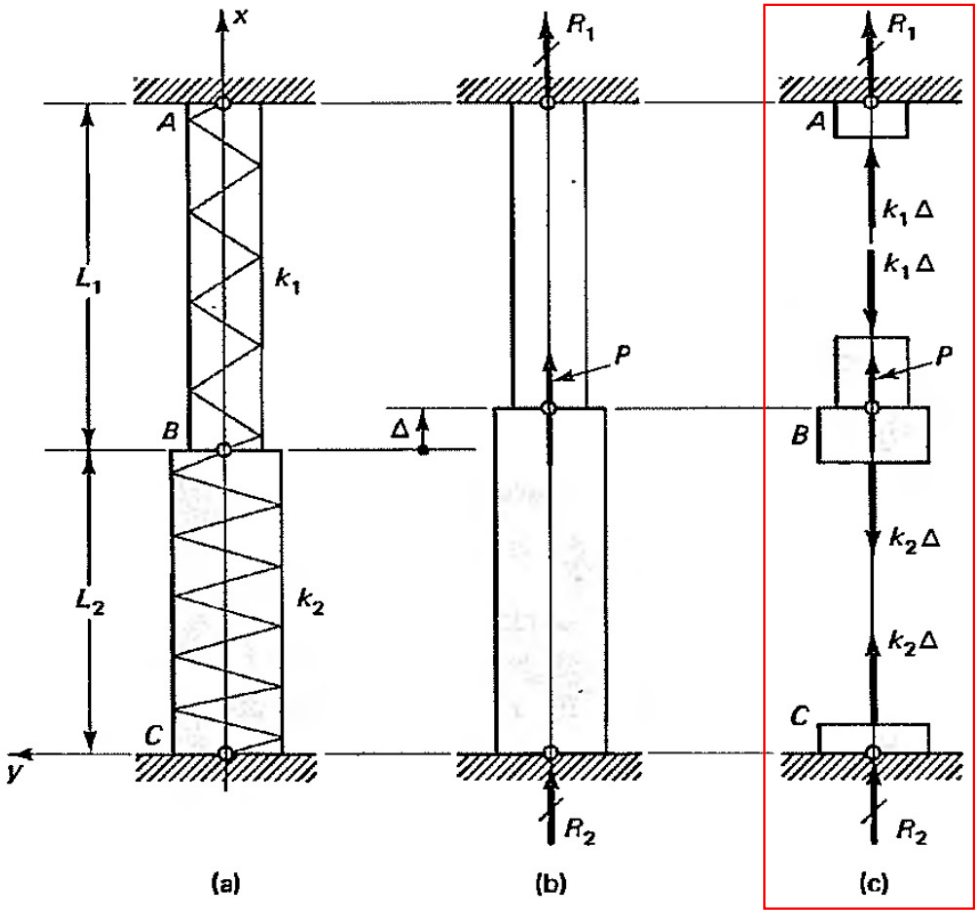
isolated free-bodies at points A,B and C. referred to as the node points.

equilibrium equation for the free body at node B

$$-k_1\Delta - k_2\Delta + P = 0$$

$$\Delta = \frac{P}{k_1 + k_2}$$

3. Displacement Method of Analysis



isolated free-bodies at points A,B and C. referred to as the node points.

equilibrium equation for the free body at node A and C.

$$R_1 = -k_1\,\Delta \quad \text{and} \quad R_2 = -k_2\,\Delta$$

$$R_1 = -\frac{k_1}{k_1 + k_2}P \qquad R_2 = -\frac{k_2}{k_1 + k_2}$$

Fig. 2-48 Displacement (stiffness) method of analysis for a statically indeterminate axially loaded bar.

3. Displacement Method - example 1

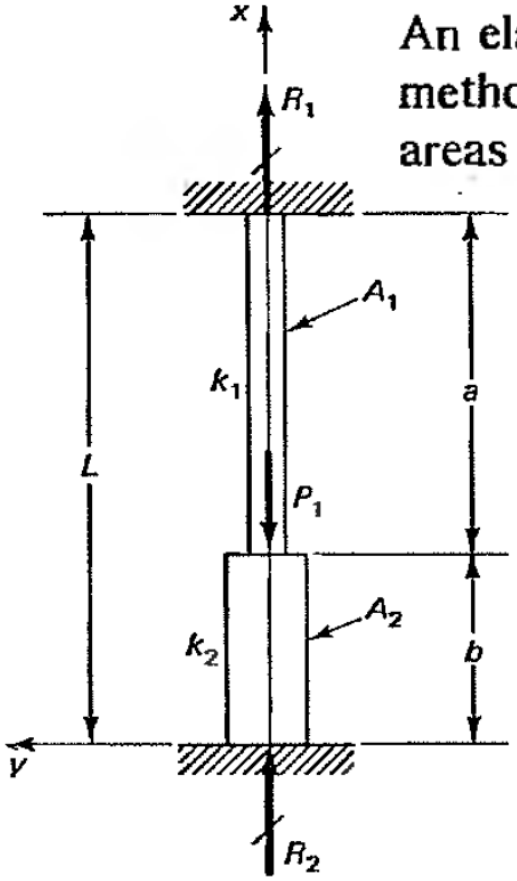


Fig. 2-49

An elastic stepped bar is loaded as shown in Fig. 2-49. Using the displacement method find the reactions. The bar segment stiffnesses k_1 and k_2, as well as their areas A_1 and A_2, and E are given.

1. isolated free-bodies at node points.
2. equilibrium equation for the free body at node points.

3. Displacement Method - several DOF

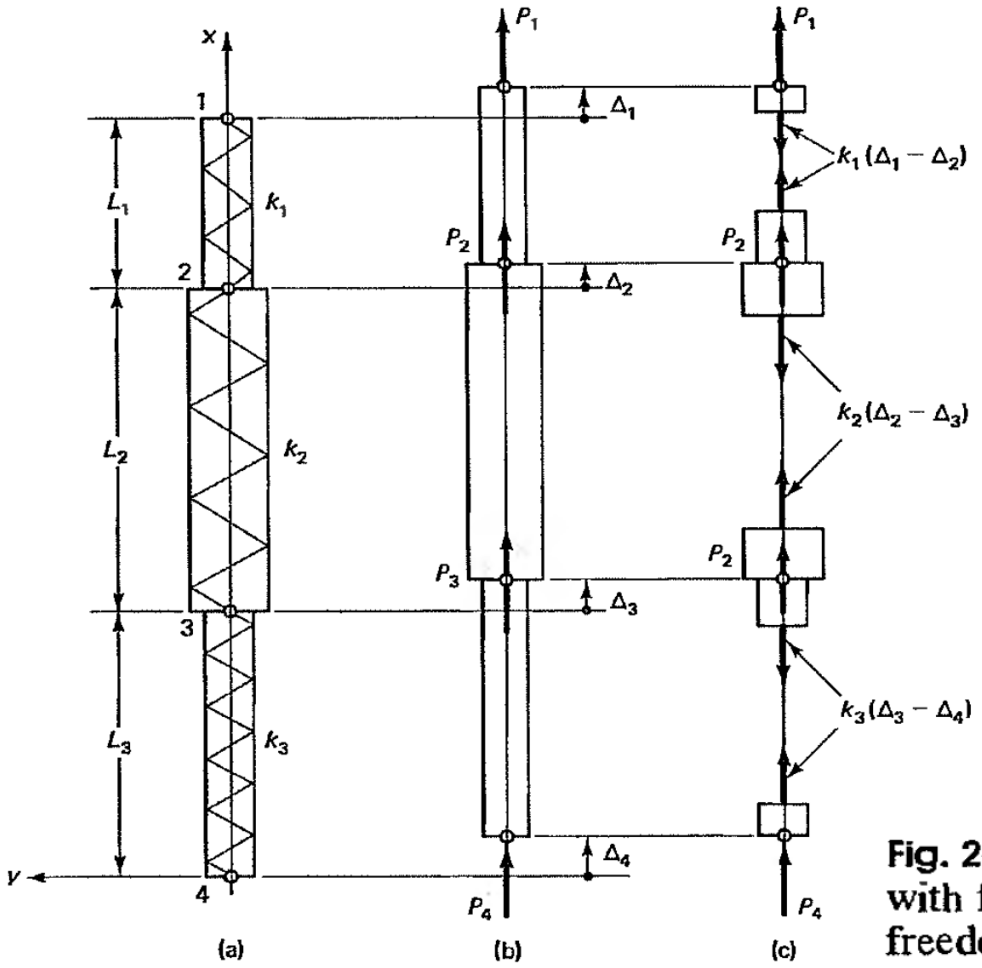
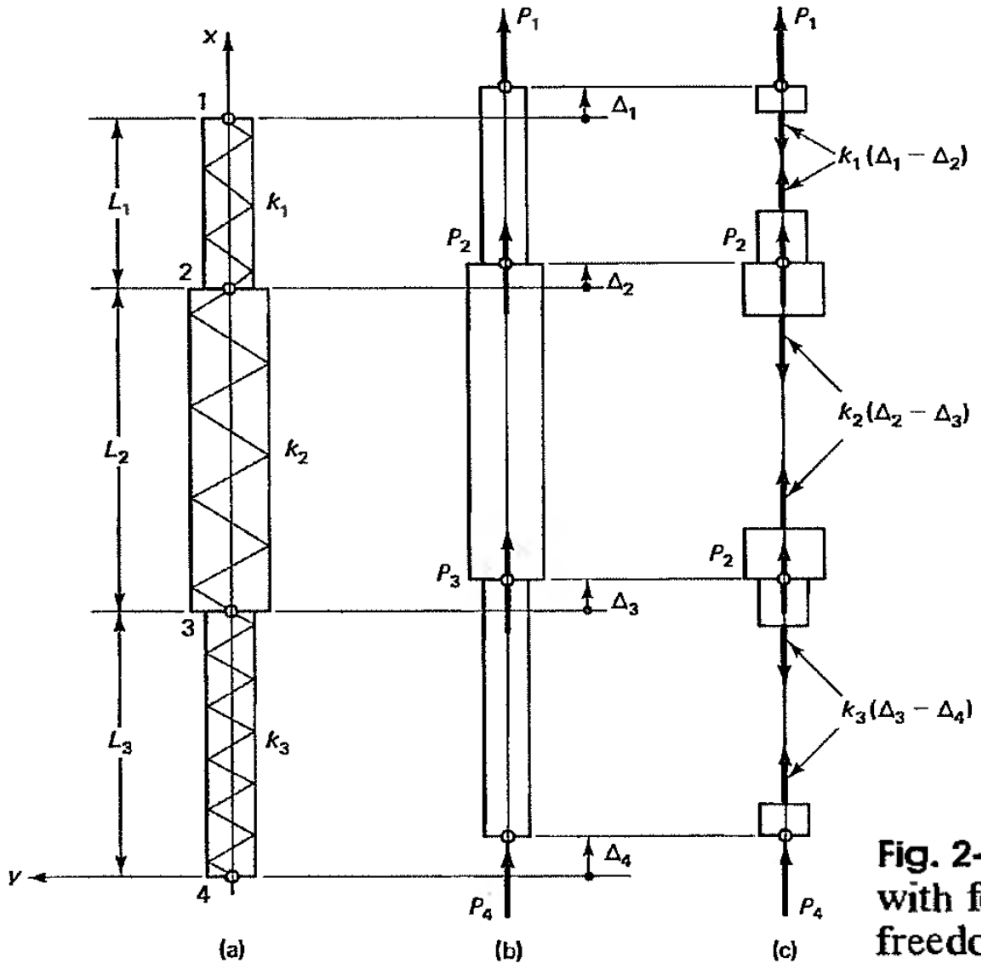


Fig. 2-50 Axially loaded bar with four degrees of freedom.

Optional, NOT Required

1. isolated free-bodies at node points, 1, 2, 3, and 4.
2. equilibrium equation for the free body at node points, 1, 2, 3, and 4.

3. Displacement Method - several DOF



$$
\begin{aligned}
P_1 - k_1(\Delta_1 - \Delta_2) &= 0 \\
P_2 + k_1(\Delta_1 - \Delta_2) - k_2(\Delta_2 - \Delta_3) &= 0 \\
P_3 + k_2(\Delta_2 - \Delta_3) - k_3(\Delta_3 - \Delta_4) &= 0 \\
P_4 + k_3(\Delta_3 - \Delta_4) &= 0
\end{aligned}
$$

$$
\begin{aligned}
k_1\Delta_1 - k_1\Delta_2 &= P_1 \\
-k_1\Delta_1 + (k_1 + k_2)\Delta_2 - k_2\Delta_3 &= P_2 \\
-k_2\Delta_2 + (k_2 + k_3)\Delta_3 - k_3\Delta_4 &= P_3 \\
-k_3\Delta_3 + k_3\Delta_4 &= P_4
\end{aligned}
$$

Fig. 2-50 Axially loaded bar with four degrees of freedom.

3. Displacement Method - several DOF

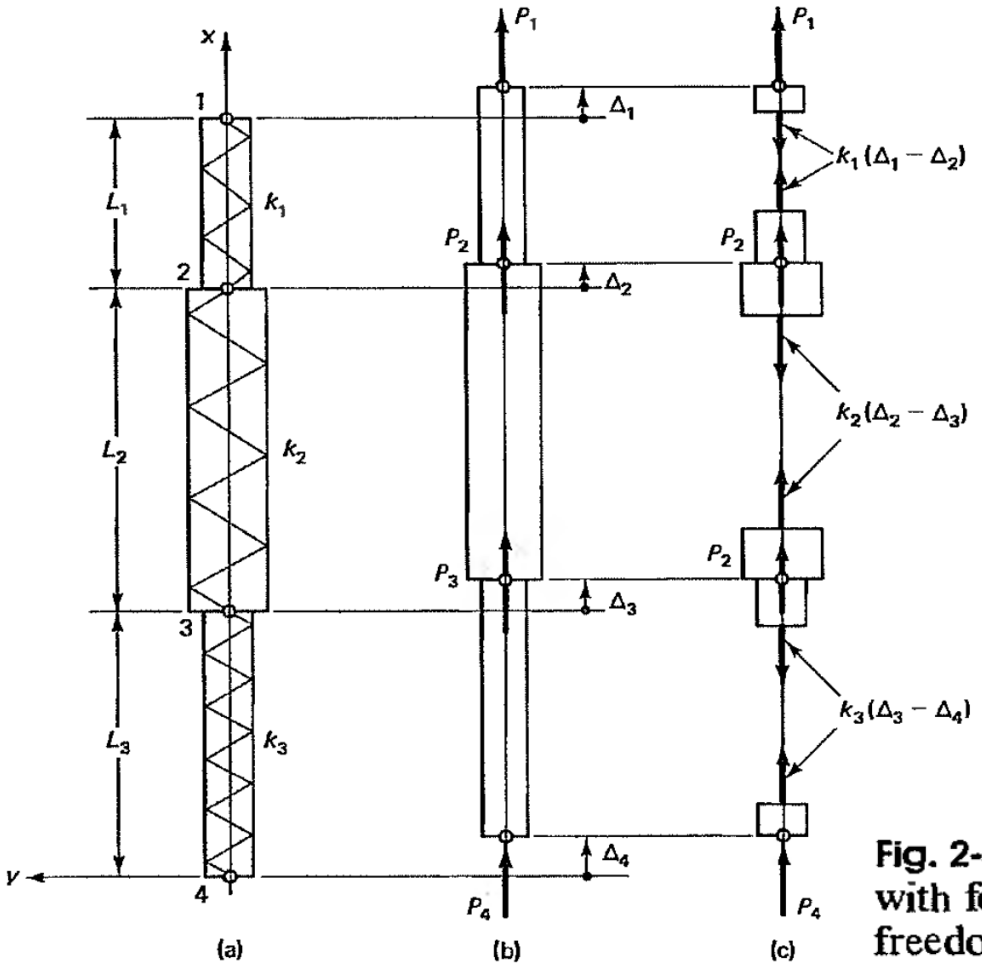


Fig. 2-50 Axially loaded bar with four degrees of freedom.

$$
\begin{aligned}
k_1\Delta_1 - k_1\Delta_2 &= P_1\\
-k_1\Delta_1 + (k_1 + k_2)\Delta_2 - k_2\Delta_3 &= P_2\\
-k_2\Delta_2 + (k_2 + k_3)\Delta_3 - k_3\Delta_4 &= P_3\\
-k_3\Delta_3 + k_3\Delta_4 &= P_4
\end{aligned}
$$

$$
\begin{bmatrix}
k_1 & -k_1 & 0 & 0\\
-k_1 & k_1 + k_2 & -k_2 & 0\\
0 & -k_2 & k_2 + k_3 & -k_3\\
0 & 0 & -k_3 & k_3
\end{bmatrix}
\begin{Bmatrix}\Delta_1\\ \Delta_2\\ \Delta_3\\ \Delta_4\end{Bmatrix}
=
\begin{Bmatrix}P_1\\ P_2\\ P_3\\ P_4\end{Bmatrix}
$$

symmetric stiffness matrix

easy to solve with computer program

3. Displacement Method - example 2

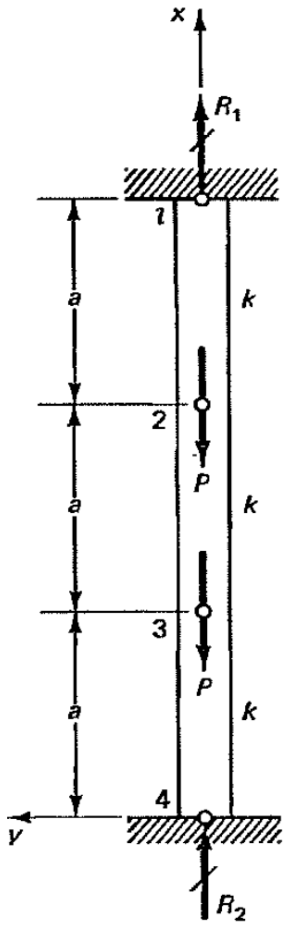


Fig. 2-51

For the elastic weightless bar held at both ends, as shown in Fig. 2-51, determine the node displacements and the reactions using the displacement method. The cross section of the bar is constant throughout.

Solution

Here only Δ_2 and Δ_3 have to be found as $\Delta_1 = \Delta_4 = 0$. Therefore, the system has two degrees of kinematic freedom. The stiffness coefficient k is the same for each segment of the bar. Applying Eqs. 2-41 and setting $\Delta_1 = \Delta_4 = 0$, one obtains

$$\begin{aligned} -k\,\Delta_2 &= R_1 \\ 2k\,\Delta_2 - k\,\Delta_3 &= -P \\ -k\,\Delta_2 + 2k\,\Delta_3 &= -P \\ -k\,\Delta_3 &= R_2 \end{aligned}$$

By solving the second and third equations simultaneously, $\Delta_2 = \Delta_3 = -P/k$, then from the first and the last equations, $R_1 = R_2 = P$. This result, which could be anticipated, means that, in effect, the upper load is hung from the top and the bottom one is supported at the base. The middle segment of the bar does not distort and deflects as a rigid body through a distance of $\Delta_2 = \Delta_3$.

In this problem, the force method would be simpler to apply than the displacement method since there is only one degree of static indeterminacy.

3. Displacement Method - example 3

(a) Consider the same loaded bar as in Example 2-16 supported only at the top and free at the bottom; see Fig. 2-52. Determine the node displacements and the reaction. For this case, $R_2 = 0$. (b) Rework part (a) if the free end is displaced $3P/k$ upwards.

Solution

(a) Here $\Delta_1 = 0$, and three nodal displacements, Δ_2, Δ_3, and Δ_4, must be determined. Therefore, this statically determinate problem has three degrees of freedom. Applying Eqs. 2-41, one has

$$\begin{aligned} -k\,\Delta_2 &= R_1 \\ 2k\,\Delta_2 - k\,\Delta_3 &= -P \\ -k\,\Delta_2 + 2k\,\Delta_3 - k\,\Delta_4 &= -P \\ -k\,\Delta_3 + k\,\Delta_4 &= 0 \end{aligned}$$

By solving the last three equations simultaneously, $\Delta_2 = 2P/k$, $\Delta_3 = \Delta_4 = -3P/k$, and then from the first equation, $R_1 = 2P$. These results can be easily checked by the procedures discussed in Part A of this chapter.

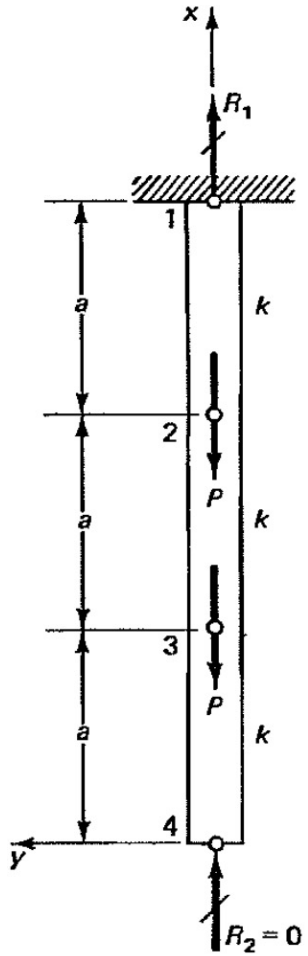


Fig. 2-52

3. Displacement Method - example 3

(a) Consider the same loaded bar as in Example 2-16 supported only at the top and free at the bottom; see Fig. 2-52. Determine the node displacements and the reaction. For this case, $R_2 = 0$. (b) Rework part (a) if the free end is displaced $3P/k$ upwards.

Solution

(b) In this case, a force R_2 of unknown magnitude must be applied at the free end to cause the specified displacement $\Delta_4 = 3P/k$. As before, $\Delta_1 = 0$. Therefore, whereas the first three equations established for part (a) apply, the fourth equation must be revised to read

$$-k\,\Delta_3 + k\,\Delta_4 = R_2$$

After substituting the given value for Δ_4 and solving the four applicable equations simultaneously, $\Delta_2 = 0$, $\Delta_3 = P/k$, and $R_1 = 0$.

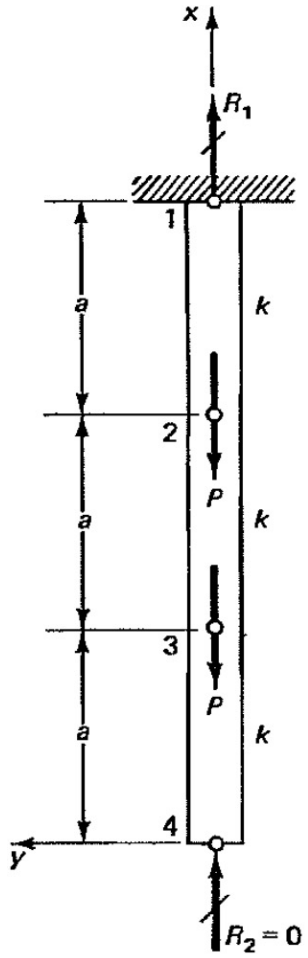


Fig. 2-52